Advanced Examples in Capital Budgeting

Example 2

Leroux Clothing Ltd is considering the acquisition of a pattern-making machine and asks for your help in evaluating this as it will only be purchased if it makes financial sense (should the proposed machine not be acquired; they would continue making the patterns manually).

The following information is supplied:

(i)	Cost of machine		R500 000
(ii)	Expected life of the machine		4 years
(iii)	Expected incremental annual accrual NET operating profits (after depreciation and interest but before tax) resulting from the operation / use of the machine:		Annual Interest expense
	Year 1	R200 000	R30 000
	Year 2	R210 000	R30 000
	Year 3	R230 000	R30 000
	Year 4	R240 000	R20 000
(iv)	The machine would require an initial working capital injection of R60 000 that would be recovered at the end of the project's life. *(raw materials required for production)*		
(v)	Leroux Clothing Ltd's cost of capital for this project is 15% per annum		
(vi)	The resale value of the machine after 4 years is expected to be R50 000.		
(vii)	Taxation rate is 28%		
(viii)	South African Revenue Services allows Section 12 C wear and tear allowance of 20 % per annum. (Also referred to as tax depreciation)		
(ix)	Extracts from interest rate tables showing the future value and present value factors using an interest rate of 15% p.a. are as follows:		
		Year	Present value of R1 at 15 %
		1	0.8696
		2	0.7561
		3	0.6575
		4	0.5718
		5	0.4972

YOU ARE REQUIRED TO:

1. Calculate the Payback Period for the machine.

2. Calculate the Net Present Value (NPV) of the proposed investment in the pattern-making machine and advise the company whether or not it should make the investment.

Example 3

New Generation Manufacturers in Soweto is considering expanding its current operations. Market research has shown that there is an increased demand for a particular that is currently not being satisfied. The firm has spent R75 000 on marketing research. The financial manager, Ms Karabo Metsing, has been tasked with purchasing a new technology enhanced machine. She prepared the following estimates and information regarding a locally manufactured machine.

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Additional information:

∴ remove ∴ 3M

1. The cost of the machine before a trade discount is R3 200 000, the supplier guarantees that the firm will receive a trade discount of R200 000 of the purchase price. Additional costs such as transport and installation costs will amount to R400 000. It is estimated that the machine will have an estimated useful life of three years and will be sold at 10 % of the original total cost at the end of three years. ∴ 3,4M

2. The machine is expected to generate the following Accrual Operating Profits for three years after deducting Depreciation expense (?) and Interest expense of R60 000 per annum: 3

	Year 1	Year 2	Year 3
Operating profits	1 800 000	2 000 000	2 400 000

3 New Generation Manufacturers will be able to claim a wear and tear allowance at 40 %, 20%, 20% in each of the three years respectively. The old machine has a tax value of nil and is expected to be sold for cash of R100 000 when the new machine arrives.

4. The firm will invest R500 000 of raw materials when the machine is ready for use. In year 2 the firm will require R550 000 and year 3, R450 000. The investment in working capital will be fully recovered at the end of the year 3. The credit terms relating to the product the new machine will produce, will be relaxed to enhance credit sales. The debtors balance relating to the credit sales at the end of year 1 will be R350 000 and year two R320 000.

5. The new machine will result in greater efficiency causing another product's sales to decline, resulting in lost profits of R80 000 in the first two years and R90 000 in year three.

6. The firm will incur fixed production costs of R650 000 to operate the machine in the first year that will increase by 5 % annually.

7. The new machine will be installed in a section of the factory that was occupied by the storeroom. The storeroom now has to be relocated to another building that will be hired at a monthly rental of R10 000. ×12 = R120 000

8. The company has a tax rate of 28 % and a conservative cost of capital of 15 %.

9. The company's expected Taxable Income for the three (3) years is expected to be as follows:

Year 1	Year 2	Year 3
770 000	1 517 500	1 533 375

Assume the numbers are correct

YOU ARE REQUIRED TO:

1. Calculate the Net Present Value (NPV) of the proposed investment in the new machine and advise the company whether or not it should make the investment. Please show all your workings.

2. Calculate whether the company expects to incur a Scrapping allowance or earn a Recoupment on the sale of the new machine at the end of the third year.

3. Name and briefly explain any three Qualitative Considerations that Ms Karabo Metsing would need to consider before advising management to invest in the new machine.

Example 4

New Horizons Ltd is considering expanding its current operations. Market research has shown that there is a demand for designer camping chairs that is currently not being satisfied. It is estimated that the demand will be for 10 000 chairs per annum and will exist for the next five years.

The company already has a factory that specialises in the production of chairs. It is estimated that modifications will need to be made to a production line that originally cost R1.2m four years ago but is currently not in use. The equipment has been fully depreciated for both tax and accounting purposes. Modifications to the production line will cost R800 000. The section 12C depreciation allowance for tax purposes of 40%; 20%; 20%; 20% per annum will apply to the modifications. If the project is not undertaken, the factory production line will be scrapped, for which a nominal amount of R120 000 will be received.

New Horizons is planning to start selling the chairs for R300 each in 2017 (year one of the project).

The market research mentioned above resulted in a cost of R100 000 which is payable on 28 February 2017.

The selling price will increase in line with inflation; this is estimated at being 4.25% pa over the life of the project. Each chair will require 3 metres of fabric and 3.5 metres of wood. The fabric costs (in 2017 terms) are R28.50 a metre and the wood R32 a metre. Each chair is expected to take 1¾ hrs to complete, and hourly wages are R22. All these costs are expected to increase at the rate of inflation.

Tax is payable at the end of each year, and the corporate tax rate is 29%.

The working capital balance at the beginning of each year is planned to be equal to 10% of that year's sales revenue.

If the designer camping chair project is not undertaken, the relevant area of the factory could be used by the salaries department. Whichever department uses the factory space will be allocated an overhead of R40 000 per annum for its use. However, if the salaries department is not accommodated in the factory, the company will incur R3 000 per month in renting external premises.

Factory overheads are R200 000 and include the depreciation of the improvements to the production line at 20% p.a. Also included in the R200 000 is the rent allocated for the factory space and the balance are fixed costs relating to the machine (these costs are not expected to increase with inflation).

It is estimated that the production line will be sold for R200 000 at the end of the project's life.

New Horizons' Weighted Average Cost of Capital (WACC) is 16%.

YOU ARE REQUIRED TO:

1. Determine whether New Horizons should undertake the new project using the NPV method. Please show all your workings

2. Calculate the project's Payback Period.